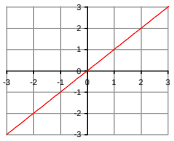
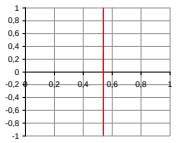
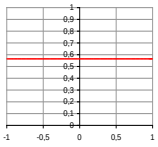
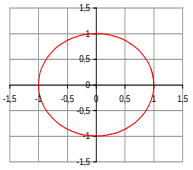
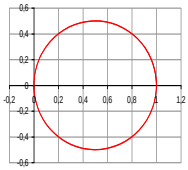
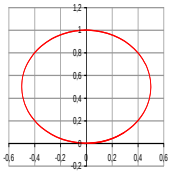
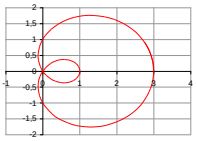
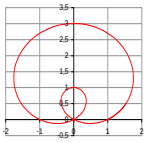
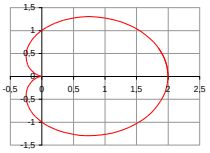
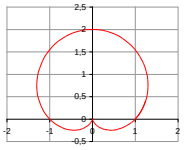
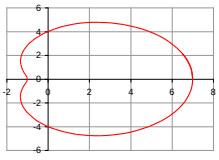
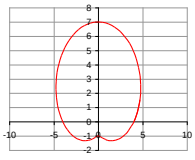
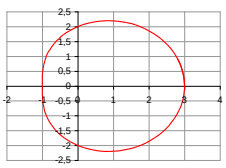
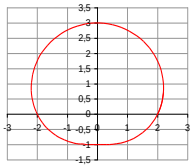
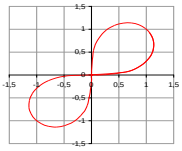
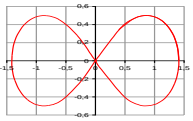
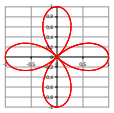
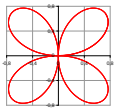
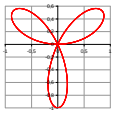
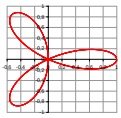
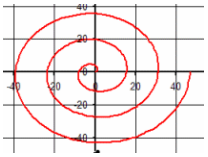
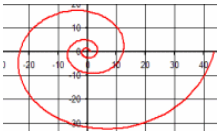
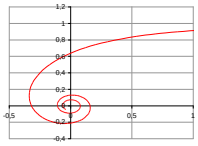
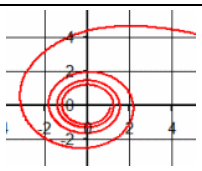
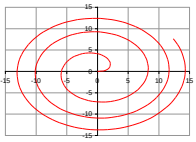
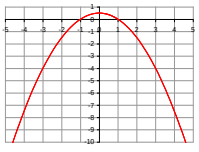
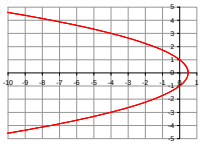
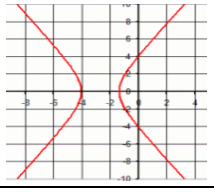
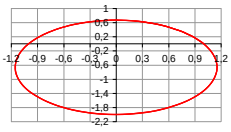
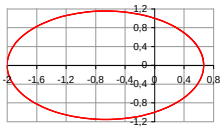
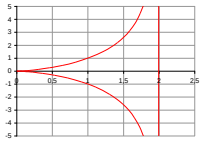
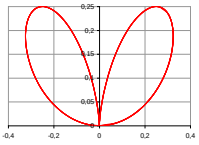
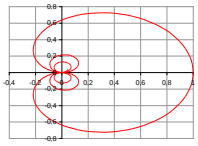
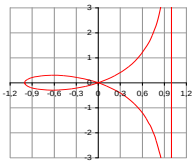
Familia de Curvas Polares más importantes		
Nombre	**Ecuación**	**Ejemplos**
Rectas:		
• por el origen	$\theta = k$	
• paralelas a un eje	$r = a\sec\theta$ $r = a\csc\theta$	 
Circunferencias:		
• con centro en el origen	$r = k$	
• tangentes a un eje	$r = a\cos\theta$ $r = asen\theta$	 
Limazones:	$\left.\begin{array}{l} r = a \pm bsen\theta \\ r = a \pm b\cos\theta \end{array}\right\} a,b > 0$	
• con rizo	$0 < \frac{a}{b} < 1$	 
• cardioide	$\frac{a}{b} = 1$	 
• con hendidura	$1 < \frac{a}{b} < 2$	 
• convexo	$\frac{a}{b} \geq 2$	 

Familia de Curvas Polares más importantes		
Nombre	**Ecuación**	**Ejemplos**
Lemniscatas	$r^2 = 2asen(2\theta)$ ó $r^2 = a^2 sen(2\theta)$ $r^2 = 2a\cos(2\theta)$ $r^2 = a^2\cos(2\theta)$	 
Rosas	$\left.\begin{array}{l} r = asen(n\theta) \\ r = a\cos(n\theta) \end{array}\right\}$ $n\ par : 2n\ pétalos$ $n\ \text{impar} : n\ pétalos$	   
Espirales:		
• de Arquímedes	$r = k\theta$	
• Logarítmica	$r = e^{a\theta}$ $r = a^{b\theta}$	
• Recíproca o Hiperbólica	$r = \frac{k}{\theta}$	
• Lituus	$r^2\theta = a^2$	
• Parabólica	$(r-a)^2 = 4ae\theta$	
Cónicas con el foco en el polo	$r = \frac{a}{1 \pm asen\theta}$ ó $r = \frac{a}{1 \pm a\cos\theta}$	
• Parábola	a = 1	 
• Hipérbola	a > 1	
• Elipse	a < 1	 

Familia de Curvas Polares más importantes		
Nombre	**Ecuación**	**Ejemplos**
Cisoide	$r = 2sen\theta \tan\theta$	
Bifolio	$r = asen\theta \cos^2\theta$	
Cocleoide	$r\theta = asen\theta$	
Estrofoide	$r = \sec\theta - 2\cos\theta$	

Trazado de curvas en Coordenadas Polares:

Una curva en Coordenadas Polares es una ecuación de la forma r = f(θ), pero:

- Algunas curvas pueden tener más de una ecuación.
- No siempre parejas equivalentes de coordenadas polares satisfacen la misma ecuación.

Los pasos para hacer una gráfica polar son:

1. Analizar simetrías:

Simetría con:	Sustitución	
Eje Polar	(r, -θ)	(-r, π – θ)
Eje π/2	(-r, -θ)	(r, π – θ)
Polo	(r, π + θ)	(-r, θ)

   Si ninguna sustitución da como resultado la ecuación original, probablemente no existe simetría.
2. Analizar si la curva es cerrada o abierta. r = f(θ) es cerrada si r no tiende a infinito para ningún valor de θ. Existen curvas abiertas con bucles.
3. Analizar puntos de corte con los ejes y con el polo:
   - Con el eje π/2, se hace θ = nπ/2 con n impar.
   - Con ele eje polar, se hace θ = nπ con n = 0, 1, 2,…
   - Con el polo, se hace r = 0 para encontrar algún θ.
4. Escoger el menor intervalo necesario para tabular, de acuerdo con los pasos anteriores.